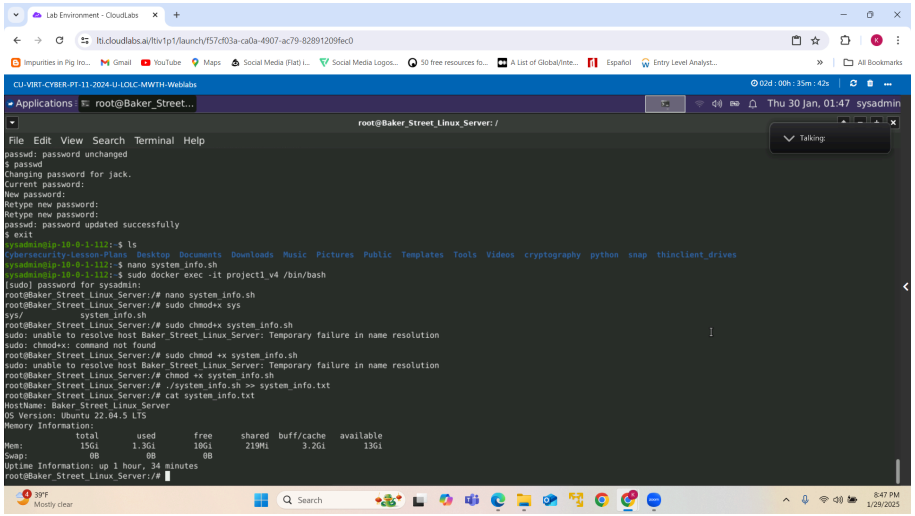
Cybersecurity

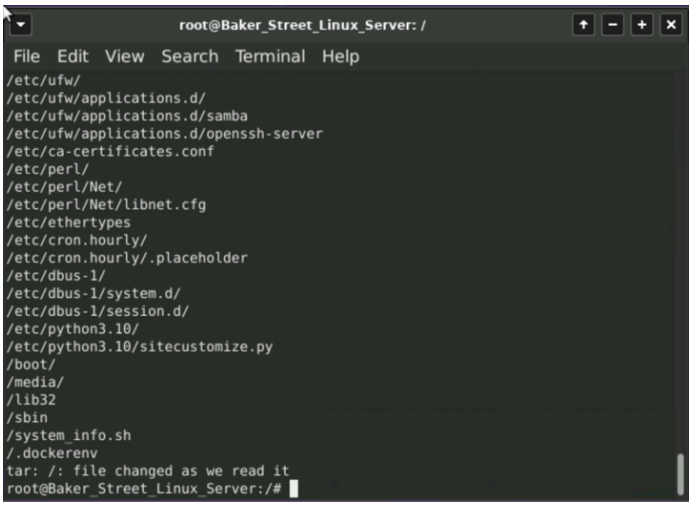
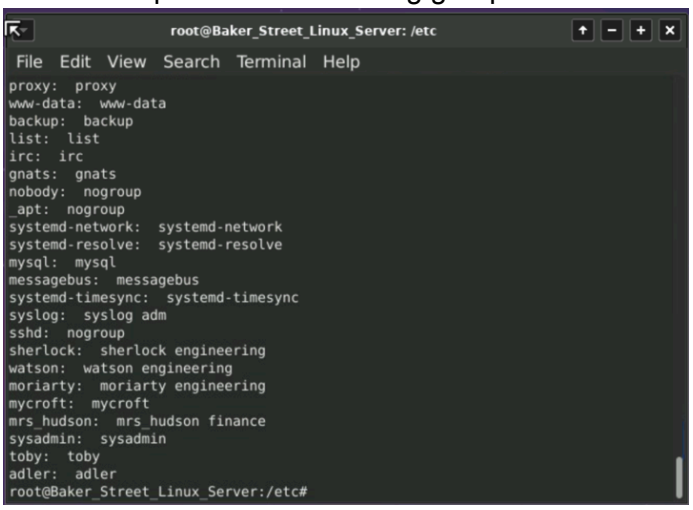
Project 1 Hardening Summary and Checklist

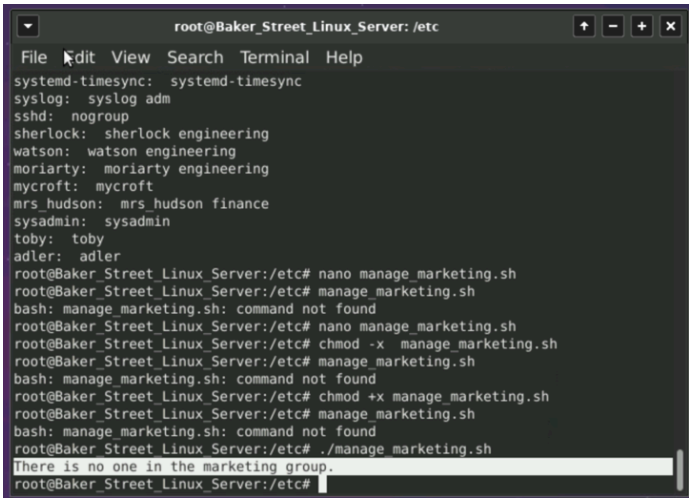
OS Information

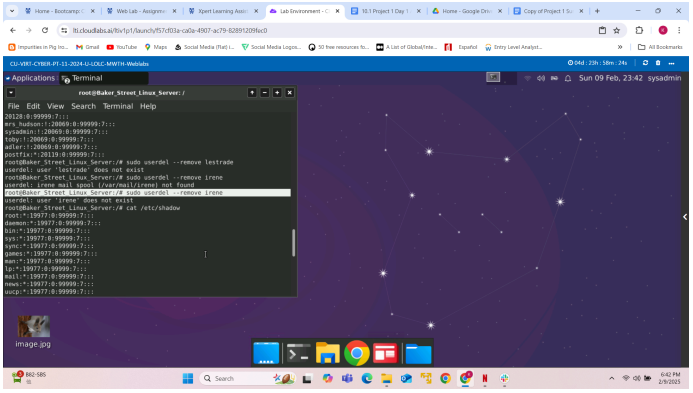
Customer	Baker Street Corporation
Hostname	Baker_Street_Linux_Server
OS Version	Ubuntu 22.04.5 LTS
Memory information	Mem: 15Gi 1.3Gi 10Gi 219Mi 3.2Gi 13Gi Swap: 0 0 0 
Uptime information	Up 1 hour, 34 minutes

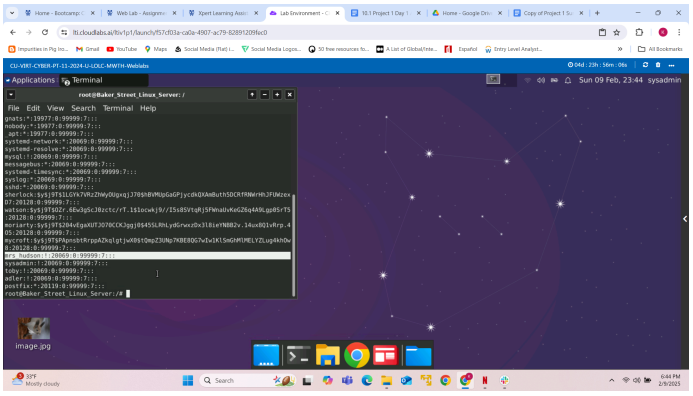
Checklist

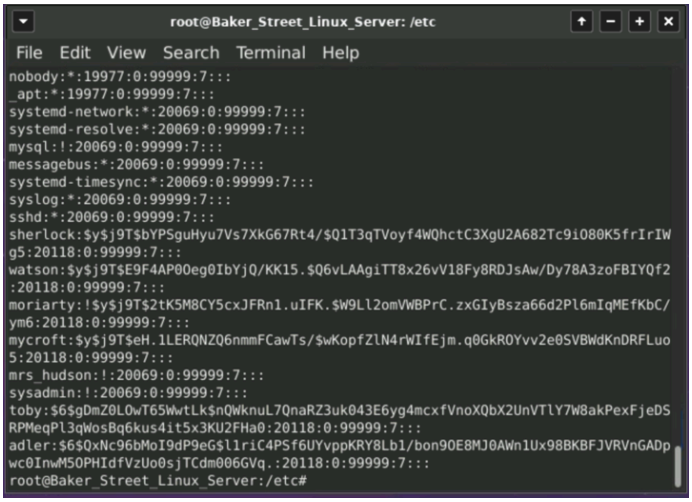
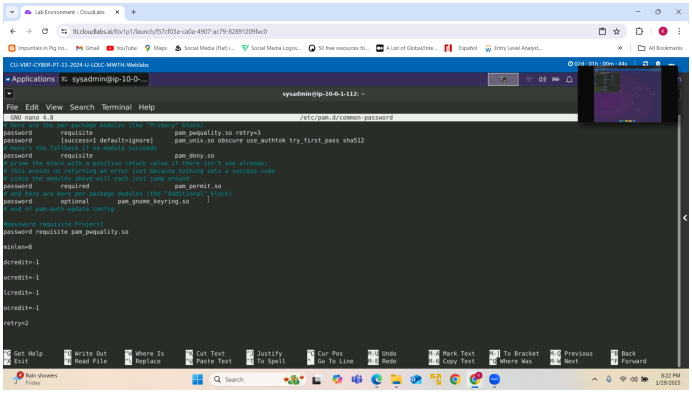
Completed	Activity	Script(s) used / Tasks completed / Screenshots

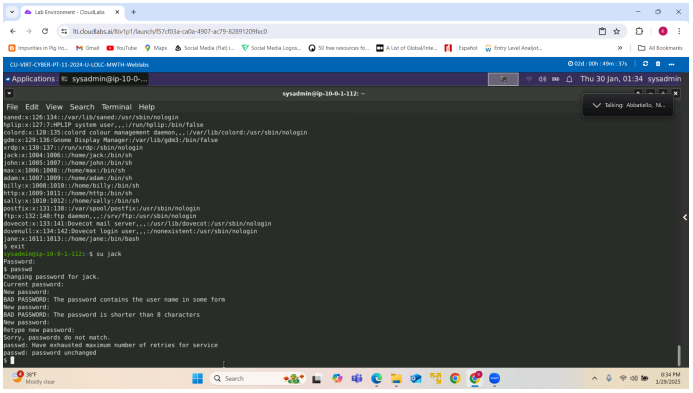
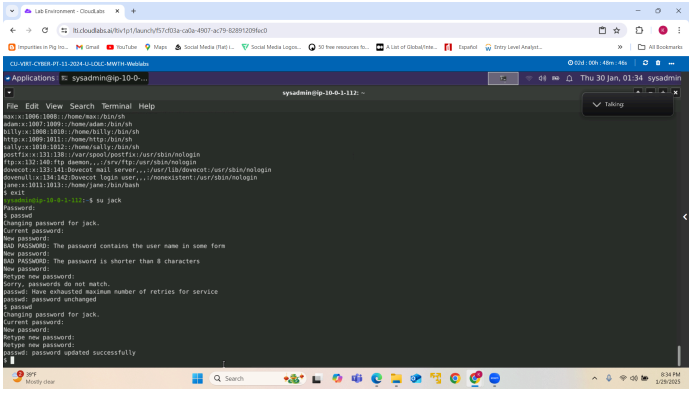
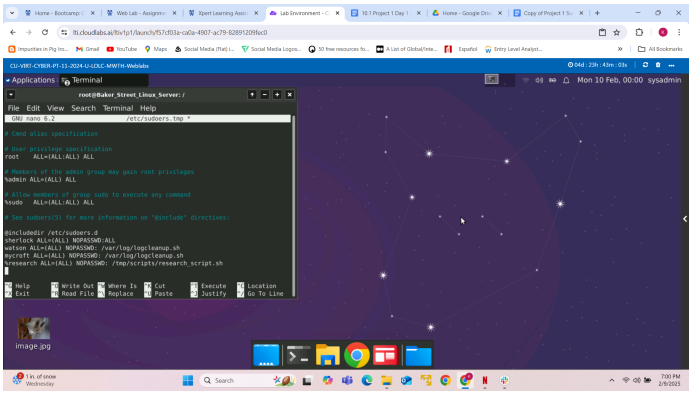
☐	OS backup	*sudo tar -cvpzf /baker_street_backup.tar.gz --exclude=/baker_street_backup.tar.gz --exclude=/proc --exclude=/tmp --exclude=/mnt --exclude=/sys --exclude=/dev --exclude=/run /* 
☐	Auditing users and groups	Using sudo userdel –remove username; remove all terminated users and sudo usermod -L username to lock all users on temporary leave. No user was found to be part of the marketing group 


root@Baker_Street_Linux_Server: /etc
File Edit View Search Terminal Help
systemd-timesync: systemd-timesync
syslog: syslog adm
sshd: nogroup
sherlock: sherlock engineering
watson: watson engineering
moriarty: moriarty engineering
mycroft: mycroft
mrs_hudson: mrs_hudson finance
sysadmin: sysadmin
toby: toby
adler: adler
root@Baker_Street_Linux_Server:/etc# nano manage_marketing.sh
root@Baker_Street_Linux_Server:/etc# manage_marketing.sh
bash: manage_marketing.sh: command not found
root@Baker_Street_Linux_Server:/etc# nano manage_marketing.sh
root@Baker_Street_Linux_Server:/etc# chmod -x manage_marketing.sh
root@Baker_Street_Linux_Server:/etc# manage_marketing.sh
bash: manage_marketing.sh: command not found
root@Baker_Street_Linux_Server:/etc# chmod +x manage_marketing.sh
root@Baker_Street_Linux_Server:/etc# manage_marketing.sh
bash: manage_marketing.sh: command not found
root@Baker_Street_Linux_Server:/etc# ./manage_marketing.sh
There is no one in the marketing group.
root@Baker_Street_Linux_Server:/etc#

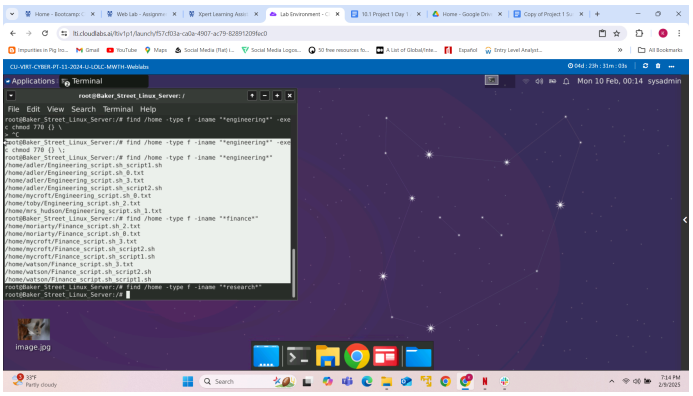
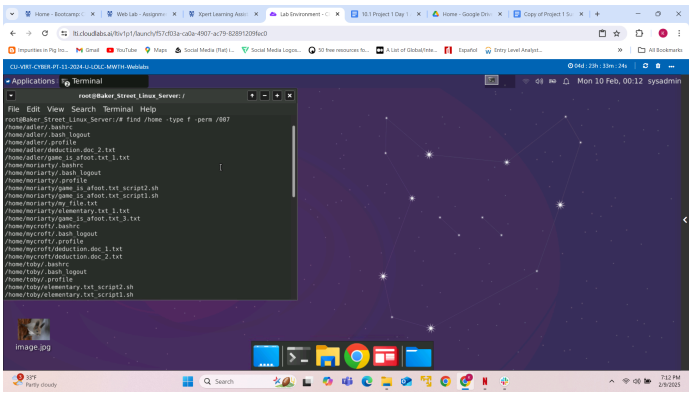

Applications
Terminal
Sun 09 Feb, 23:42
sysadmin
image.jpg

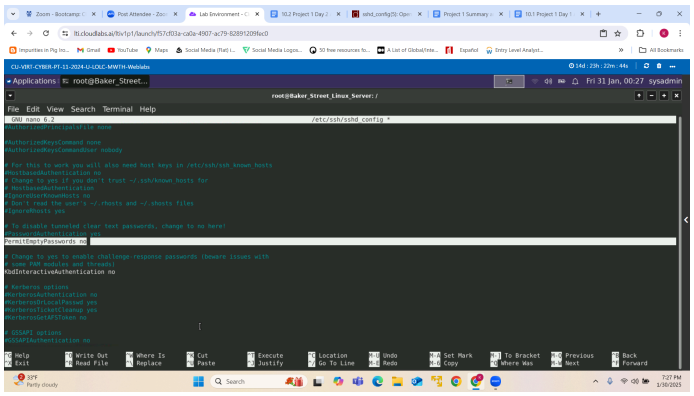
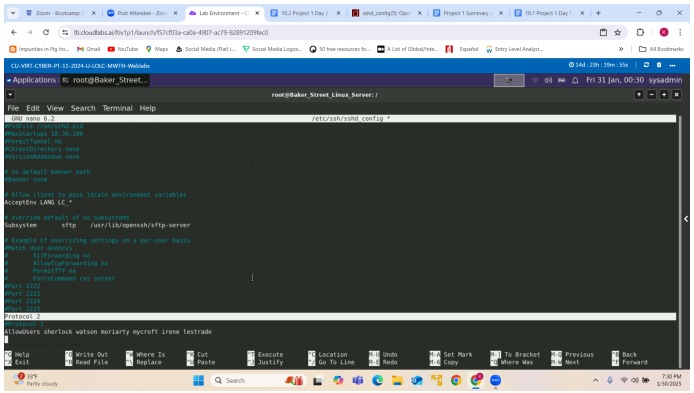
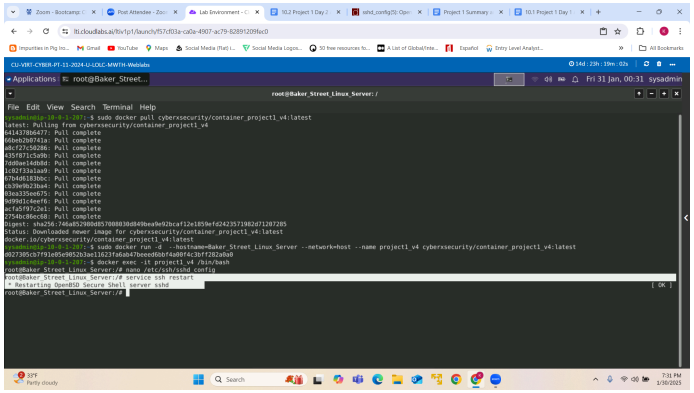

Applications
Terminal
Sun 09 Feb, 23:44
sysadmin
image.jpg

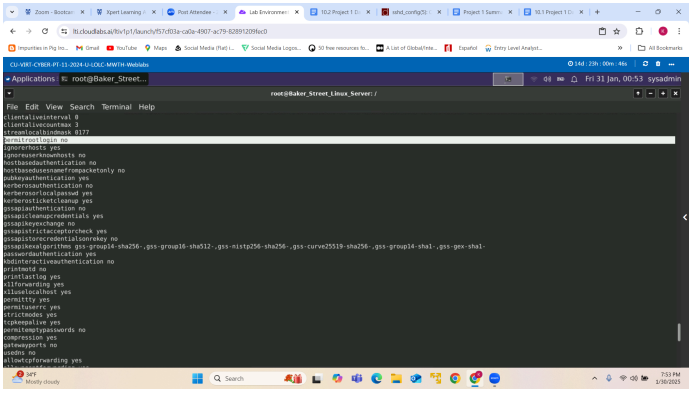
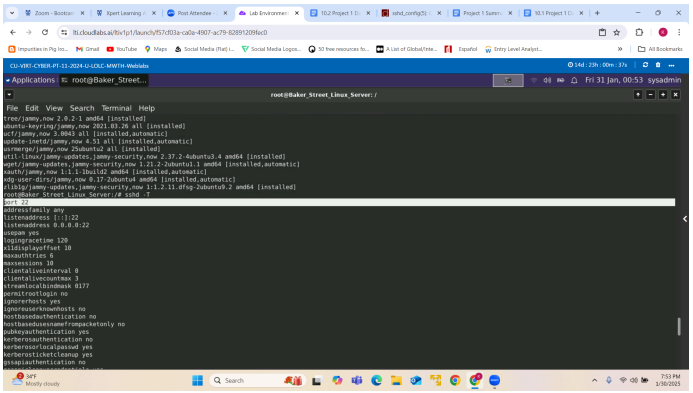
		
☐	Updating and enforcing password policies	Using *sudo apt install libpam-pwquality* To set password requirements for users and using User Jack to test new changes. 

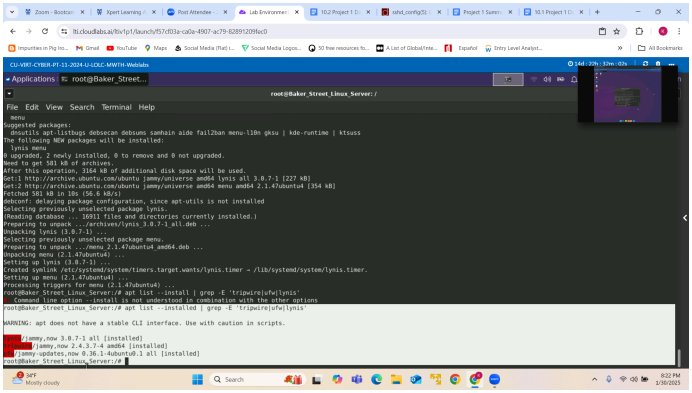
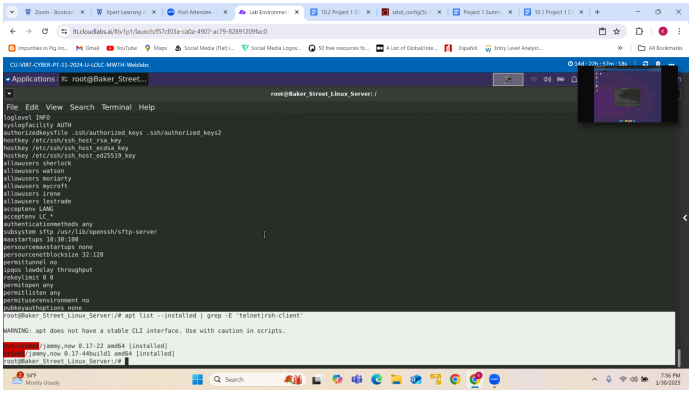
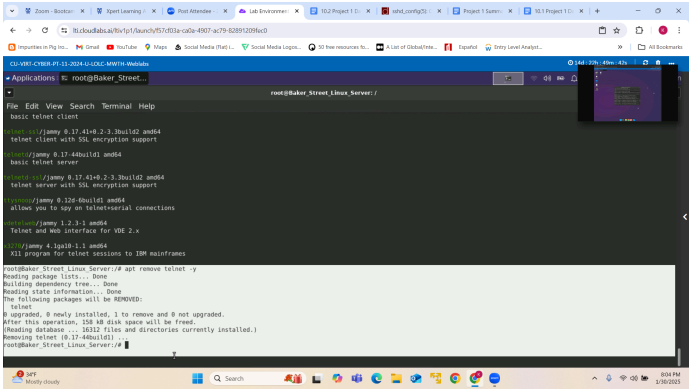
		 
☐	Updating and enforcing sudo permissions	Sudo visudo to edit the sudoers files 

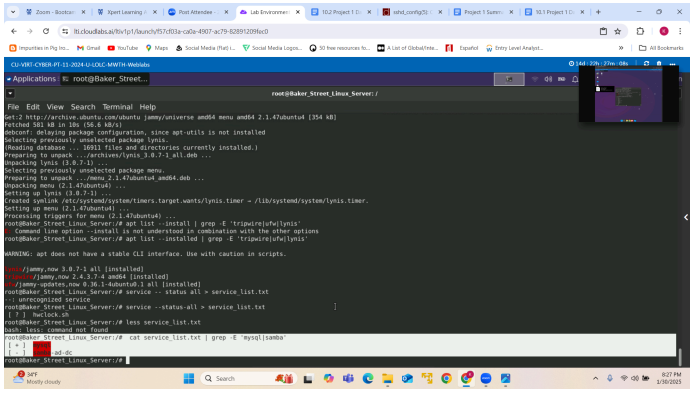
☐	Validating and updating permissions on files and directories	`find /home -type f -perm /007`
		`find /home -type f -perm /007 -exec chmod o-rwx {} \;`
		`find /home -type f -iname "*finance*" -exec chmod 770 {} \`
		
		`find /home -type f -iname "*engineering*" -exec chmod 770 {} \`

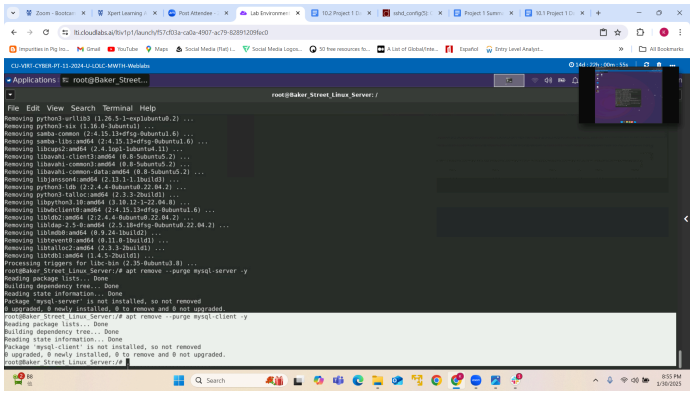
		`find /home -type f -iname "*research*" -exec chmod 770 {} \;`  
☐	Optional: Updating password hashing configuration	

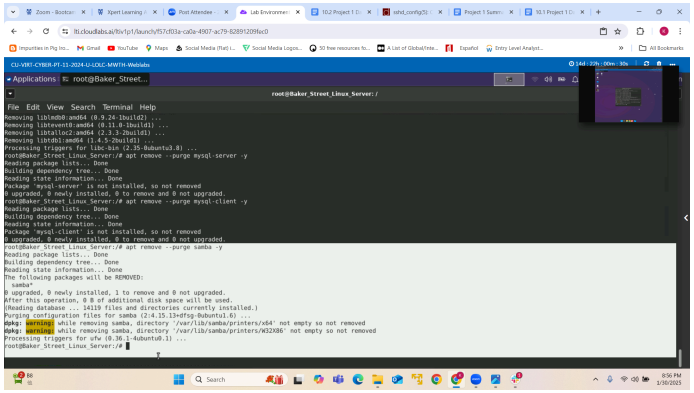
☐	Auditing and securing SSH	nano /etc/ssh/ssh_config   

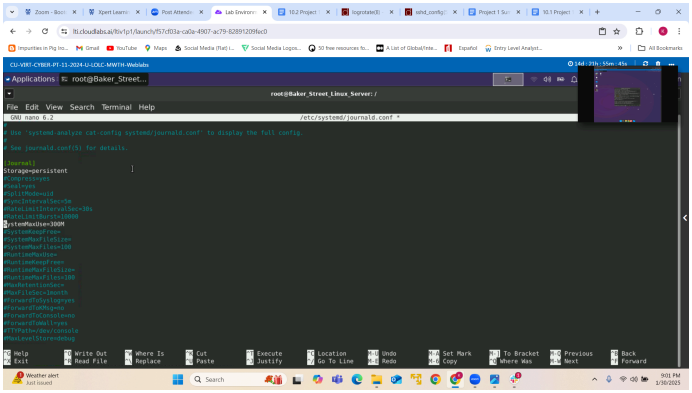
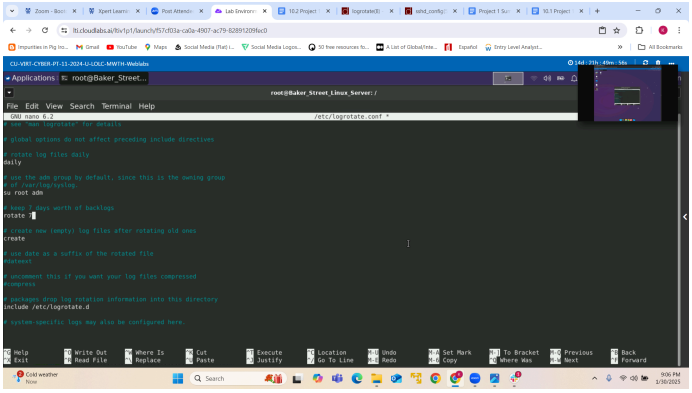
		 
☐	Reviewing and updating system packages	apt update apt upgrade -y apt list –installed >> package_list.txt apt autoremove -y Removed servicesTelnet & rsh-client Installed packages lyons, ufw and tripwire. Tripwire offers several hardening features that enhance the security of systems by monitoring and managing file integrity. Uncomplicated Firewall (UFW) is a user-friendly interface for managing iptables firewall rules on Linux systems. It

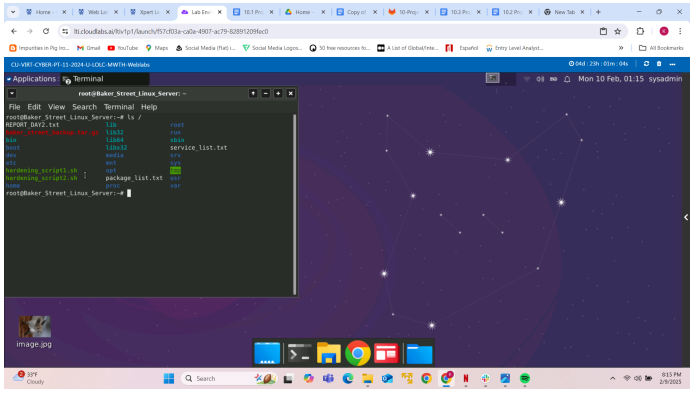
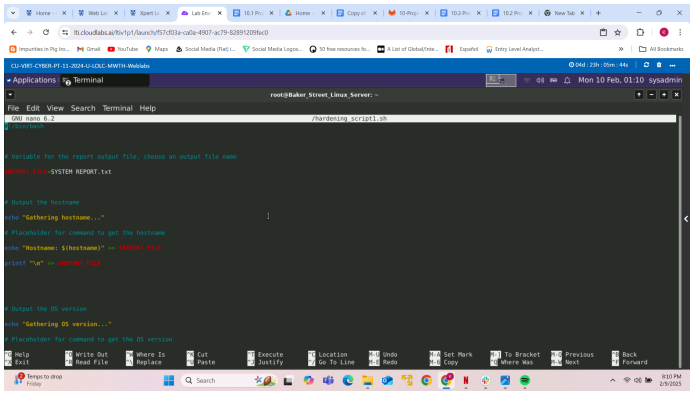
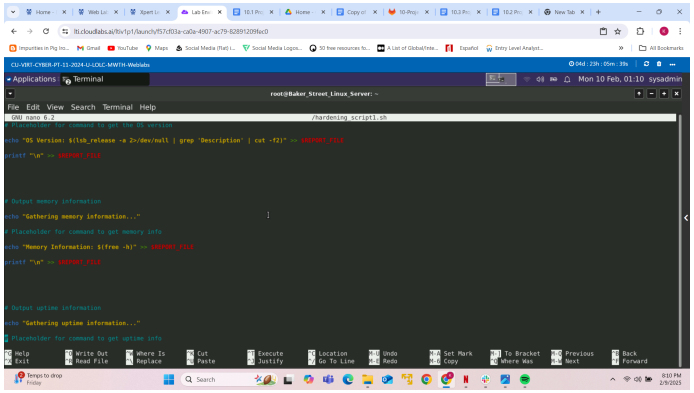
		provides several hardening features to enhance the security of your system. Lynis is a security auditing tool for Unix-based systems that helps assess the security posture of a system and provides recommendations for hardening it.  
☐	Disabling unnecessary services	

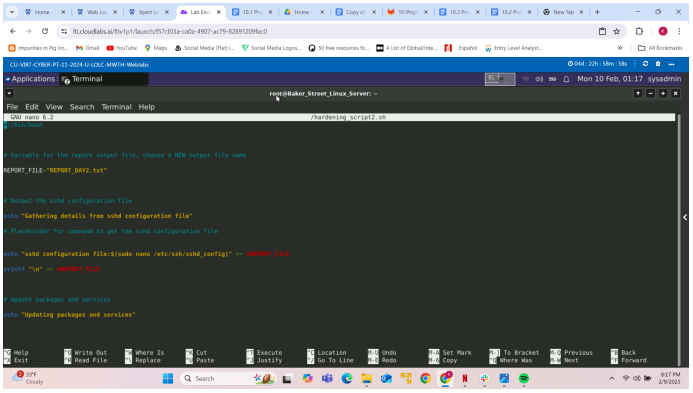

root@Baker_Street_Linux_Server: /
File Edit View Search Terminal Help
Get:2 http://archive.ubuntu.com/ubuntu jammy/universe amd64 menu amd64 2.1.47ubuntu4 [354 kB]
Fetched 501 kB in 10s (56.6 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynis.
(Reading database ... 16911 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.7-1_all.deb ...
Unpacking lynis (3.0.7-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../menu_2.1.47ubuntu4_amd64.deb ...
Unpacking menu (2.1.47ubuntu4) ...
Setting up lynis (3.0.7-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /lib/systemd/system/lynis.timer.
Setting up menu (2.1.47ubuntu4) ...
Processing triggers for menu (2.1.47ubuntu4) ...
root@Baker_Street_Linux_Server:/# apt list --install | grep -E 'tripwire|ufw|lynis'
E: Command line option --install is not understood in combination with the other options
root@Baker_Street_Linux_Server:/# apt list --installed | grep -E 'tripwire|ufw|lynis'
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
lynis/jammy,now 3.0.7-1 all [installed]
tripwire/jammy,now 2.4.3.7-4 amd64 [installed]
ufw/jammy-updates,now 0.36.1-4ubuntu0.1 all [installed]
root@Baker_Street_Linux_Server:/# service -- status all > service_list.txt
-: unrecognized service
root@Baker_Street_Linux_Server:/# service --status-all > service_list.txt
[?] hwclock.sh
root@Baker_Street_Linux_Server:/# less service_list.txt
bash: less: command not found
root@Baker_Street_Linux_Server:/# cat service_list.txt | grep -E 'mysql|samba'
[+] smbd
[-] samba-ad-dc
root@Baker_Street_Linux_Server:/#

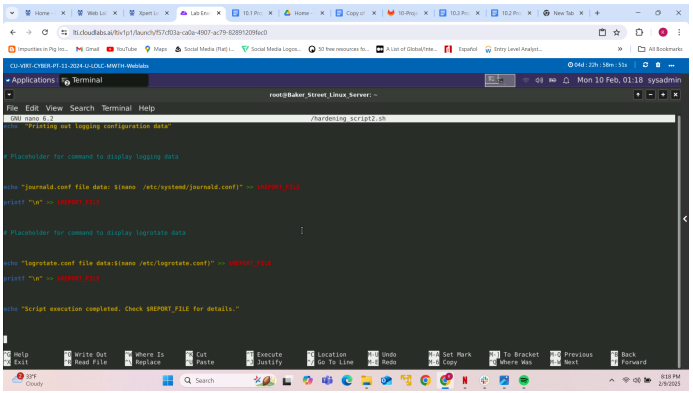

root@Baker_Street_Linux_Server: /
File Edit View Search Terminal Help
Removing python3-urllib3 (1.26.5-1~exp1ubuntu0.2) ...
Removing python3-six (1.16.0-3ubuntu1) ...
Removing samba-common (2:4.15.13+dfsg-0ubuntu1.6) ...
Removing samba-libs:amd64 (2:4.15.13+dfsg-0ubuntu1.6) ...
Removing libcups2:amd64 (2.4.1op1-1ubuntu4.21) ...
Removing libavahi-client3:amd64 (0.8-5ubuntu5.2) ...
Removing libavahi-common3:amd64 (0.8-5ubuntu5.2) ...
Removing libavahi-common-data:amd64 (0.8-5ubuntu5.2) ...
Removing libjansson4:amd64 (2.13.1-1.1build3) ...
Removing python3-ldb (2:2.4.4-0ubuntu0.22.04.2) ...
Removing python3-talloc:amd64 (2.3.3-2build1) ...
Removing libpython3.10:amd64 (3.10.12-1~22.04.8) ...
Removing libwbclient0:amd64 (2:4.15.13+dfsg-0ubuntu1.6) ...
Removing libtdb2:amd64 (2:2.4.6-0ubuntu0.22.04.2) ...
Removing libldap-2.5-0:amd64 (2.5.18+dfsg-0ubuntu0.22.04.2) ...
Removing libtdb0:amd64 (0.9.24-1build2) ...
Removing libtevent0:amd64 (0.11.0-1build1) ...
Removing libtalloc2:amd64 (2.3.3-2build1) ...
Removing libtdb1:amd64 (1.4.5-2build1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
root@Baker_Street_Linux_Server:/# apt remove --purge mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'mysql-server' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Baker_Street_Linux_Server:/# apt remove --purge mysql-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'mysql-client' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Baker_Street_Linux_Server:/#

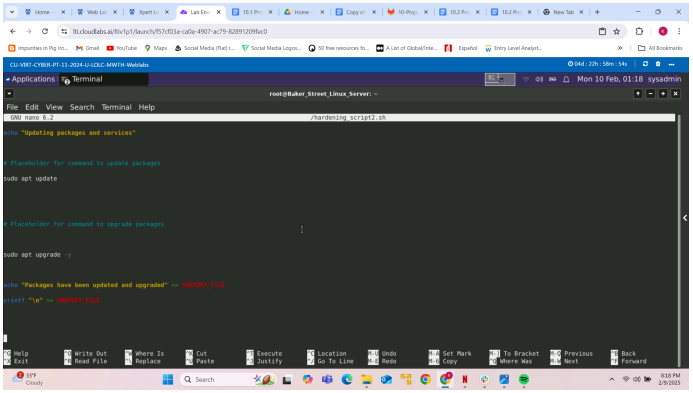

root@Baker_Street_Linux_Server: /
File Edit View Search Terminal Help
Removing libtdb0:amd64 (0.9.24-1build2) ...
Removing libtevent0:amd64 (0.11.0-1build1) ...
Removing libtalloc2:amd64 (2.3.3-2build1) ...
Removing libtdb1:amd64 (1.4.5-2build1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
root@Baker_Street_Linux_Server:/# apt remove --purge mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'mysql-server' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Baker_Street_Linux_Server:/# apt remove --purge mysql-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'mysql-client' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Baker_Street_Linux_Server:/# apt remove --purge samba -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  samba*
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 0 B of additional disk space will be used.
(Reading database ... 14119 files and directories currently installed.)
Purging configuration files for samba (2:4.15.13+dfsg-0ubuntu1.6) ...
dpkg: warning: while removing samba, directory '/var/lib/samba/printers/x64' not empty so not removed
dpkg: warning: while removing samba, directory '/var/lib/samba/printers/W32X86' not empty so not removed
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
root@Baker_Street_Linux_Server:/#

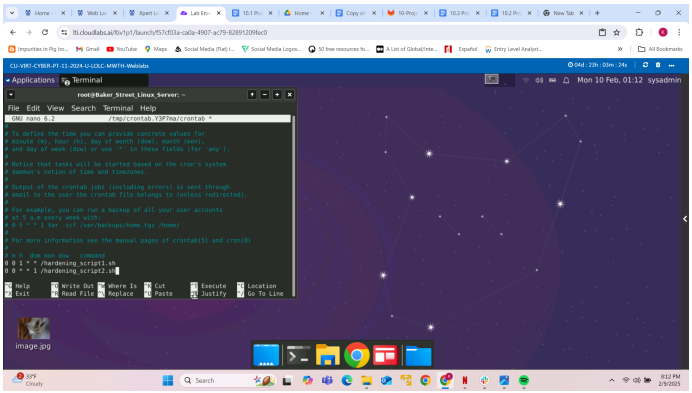
☐	Enabling and configuring logging	Nano /etc/logrotate.conf nano /etc/systemd/journald.conf  

☐	Scripts created	  


root@Baker_Street_Linux_Server: ~
GNU nano 6.2 /hardening_script2.sh
Variable for the report output file, choose a NEW output file name
REPORT_FILE="REPORT_DAY2.txt"
Output the sshd configuration file
echo "Gathering details from sshd configuration file"
Placeholder for command to get the sshd configuration file
echo "sshd configuration file:$(sudo nano /etc/ssh/sshd_config)" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE
Update packages and services
echo "Updating packages and services"


root@Baker_Street_Linux_Server: ~
GNU nano 6.2 /hardening_script2.sh
echo "Printing out logging configuration data"
Placeholder for command to display logging data
echo "journald.conf file data: $(nano /etc/systemd/journald.conf)" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE
Placeholder for command to display logrotate data
echo "logrotate.conf file data:$(nano /etc/logrotate.conf)" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE
echo "Script execution completed. Check $REPORT_FILE for details."


root@Baker_Street_Linux_Server: ~
GNU nano 6.2 /hardening_script2.sh
echo "Updating packages and services"
Placeholder for command to update packages
sudo apt update
Placeholder for command to upgrade packages
sudo apt upgrade -y
echo "Packages have been updated and upgraded" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

☐	Scripts scheduled with cron	Crontab -e 

Conclusion:

- System password policies needed updating and enforcing to better protect user accounts and make it harder for intruders to gain access using accounts on the system.
- Services with known vulnerabilities were removed and security packages were added to further harden the system and improve integrity, auditing of the system.
- Users, groups, files and directories were audited to give access to only what groups and users require.